

Factors to be considered while choosing the right database for your application

Well, there is a difference between choosing any database for an application and choosing the right database for it. Let's have a look at some of the important factors to be considered while choosing a database for an application.

Structure of data: The structure of the data basically decides how we need to store and retrieve it. As our applications deal with data present in a variety of formats, selecting the right database should include picking the right data structures for storing and retrieving the data. If we do not select the right data structures for persisting our data, our application will take more time to retrieve data from the database, and will also require more development efforts to work around any data issues.

Size of data to be stored: This factor takes into consideration the quantity of data we need to store and retrieve as critical application data. The amount of data we can store and retrieve may vary depending on a combination of the data structure selected, the ability of the database to differentiate data across multiple file systems and servers, and even vendor-specific optimisations. So we need to choose our database keeping in mind the overall volume of data generated by the application at any specific time and also the size of data to be retrieved from the database.

Speed and scalability: This decides the speed we require for reading the data from the database and writing the data to the database. It addresses the time taken to service all incoming reads and writes to our application. Some databases are actually designed to optimise read-heavy applications, while others are designed in a way to support write-heavy solutions. Selecting a database that can handle our application's input/output needs can actually go a long way to making a scalable architecture.

Accessibility of data: The number of people or users concurrently accessing the database and the level of computation involved in accessing any specific data are also important factors to consider while choosing the right database. The processing speed of the application gets affected if the database chosen is not good enough to handle large loads.

Data modelling: This helps map our application's features into the data structure and we will need to implement the same. Starting with a conceptual model, we can identify the entities, their associated attributes, and the entity relationships that we will need. As we go through the process, the type of data structures we will need in order to implement the application will become more apparent. We can then use these structural considerations to select the right category of database that will serve our application the best.

Scope for multiple databases: During the modelling process, we may realise that we need to store our data in a specific data structure, where certain queries cannot

be optimised fully. This may be because of various reasons such as some complex search requirements, the need for robust reporting capabilities, or the requirement for a data pipeline to accept and analyse the incoming data. In all such situations, more than one type of database may be required for our application. When choosing more than one database, it's quite important to select one database that will own any specific set of data. This database acts as the canonical database for those entities. Any additional databases that work with this same set of data may have a copy, but will not be considered as the owner of this data.

Safety and security of data: We should also check the level of security that any database provides to the data stored in it. In scenarios where the data to be stored is highly confidential, we need to have a highly secured database. The safety measures implemented by the database in case of any system crash or failure is quite a significant factor to keep in mind while choosing a database.

A few open source database solutions available in the market

MySQL

MySQL has been around since 1995 and is now owned by Oracle. Apart from its open source version, there are also different paid editions available that offer some additional features, like automatic scaling and cluster geo-replication. We know that MySQL is an industry standard now, as it's compatible with just about every operating system and is written in both C and C++. This database solution is a great option for different international users, as the server can provide different error messages to clients in multiple languages, encompassing support for several different character sets.

Pros

- It can be used even when there is no network available.
- It has a flexible privilege and password system.
- It uses host-based verification.
- It has security encryption for all the password traffic.
- It consists of libraries that can be embedded into different standalone applications.

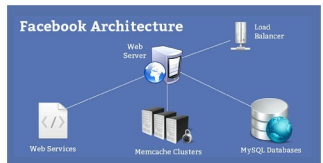


Figure 3: Facebook architecture using MySQL as the database
(Image source: googleimages.com)